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Please write clearly in block capitals.

Centre number | | | | | |

Candidate number | | | | |

Surname

Forename(s)

Candidate signature

Computer Science

Paper 1 topics: Computer systems (1.1 to 1.6)

Suggested time: about 30 minutes. There is no timer.

Materials

For this paper you must have:

- a pen
- no calculator is allowed

Instructions

- Use black ink.
- Write your answer to each question in the space provided.
- Answer all the questions.
- Do not use a calculator.

Information

- The total mark for this paper is 30.
- The marks for each question are shown in brackets [ ].

For Examiner's Use	
Question	**Mark**
1	
2	
3	
4	
5	
TOTAL	

This is a practice paper written for Dougie in the style of the GCSE exam. It is not an exam board paper, and it is not a paper from any past exam series.

1 **(a)** State the purpose of the CPU.

[1]

(b) Which register holds the address of the next instruction to be fetched? Tick (✓) one box.

Accumulator ☐

Memory Address Register (MAR) ☐

Memory Data Register (MDR) ☐

Program Counter (PC) ☐

[1]

(c) Explain how increasing the number of cores in a CPU can improve its performance.

[2]

(d) Give one example of an embedded system.

[1]

2 **(a)** Explain two differences between RAM and ROM.

[2]

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(b) Convert the binary number 10110110 into denary.

[1]

(c) Convert the denary number 75 into an 8-bit binary number.

[1]

(d) Convert the hexadecimal number 3F into denary.

[1]

(e) Add the binary numbers 01101100 and 00110101. Give your answer as an 8-bit binary number. Show your working.

[2]

(f) An image is 100 pixels wide and 50 pixels high. It has a colour depth of 8 bits. Calculate the file size of the image in bytes. Ignore metadata. Show your working.

[2]

(g) Describe one difference between lossy and lossless compression.

[2]

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3 **(a)** State what is meant by a LAN.

[1]

(b) A school uses a star topology. Explain one advantage of a star topology.

[2]

(c) Which protocol is used to send emails? Tick (✓) one box.

FTP ☐

HTTP ☐

IMAP ☐

SMTP ☐

[1]

(d) Describe what a DNS (Domain Name Server) does when a user types a web address into a browser.

[2]

4 **(a)** Describe what is meant by phishing.

[2]

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(b) Explain how a firewall helps to protect a network.

[2]

(c) Identify one function of an operating system and describe what it does.

[2]

5 State two principles of the Data Protection Act 2018 that an online shop must follow with its customers' data.

[2]

END OF QUESTION PAPER

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